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TERM END EXAMINATIONS (TEE) – April 2026

Programme	:	**B.Tech.**	**Semester**	:	**Winter Semester 2025-26**
Course Title	:	**Computer Architecture and Organization**	**Course Code**	:	**CSE2003**
Date/ Session	:	**21st April 2026 / Session-I**	**Slot**	:	**B21+E14+E22**
Time	:	**3 Hrs.**	**Max. Marks**	:	**100**

Answer ALL the Questions

Q. No.		Question Description	Marks
		PART A – (60 Marks)	
1	(a)	Illustrate an instruction set. Discuss CISC and RISC.	12
		OR	
	(b)	Briefly discuss following address modes: i. Immediate Addressing Mode ii. Direct Memory Addressing Mode iii. Indirect Memory Addressing Mode iv. Direct Register Addressing Mode v. Indirect Register Addressing Mode vi. Stack Addressing Mode	12
2	(a)	Perform the number conversions for the following numbers: i. $(567)_8 \rightarrow (\ )_2$ ii. $(AF45)_H \rightarrow (\ )_{10}$ iii. $(821.58)_{10} \rightarrow (\ )_2$ iv. $(11011.1010)_2 \rightarrow (\ )_{10}$	12
		OR	
	(b)	Find out the multiplication of the followings: i. $(3 \times 4)_{10} \rightarrow (\ )_2$ ii. $(-6 \times -5)_{10} \rightarrow (\ )_2$	12
3	(a)	Consider a **direct mapped cache** of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find- i. Number of bits in tag ii. Tag directory size	12
		OR	
	(b)	Consider a **2-way set associative mapped cache** of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find- i. Number of bits in tag ii. Tag directory size	12
4	(a)	Differentiate secondary memory with primary memory. Discuss briefly the followings:	12

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		i. Magnetic Disk ii. CD	
		OR	
	(b)	Illustrate RAID.	12
5	(a)	Define Handshaking. Briefly discuss the handshaking process.	12
		OR	
	(b)	Illustrate interrupts.	12
		PART B – (40 Marks)	
6		Differentiate the Von Neumann Machine and Harvard Architecture.	8
7		Consider a direct mapped cache of size 512 KB with block size 1 KB. There are 7 bits in the tag. Find-Size of main memory.	8
8		Consider a fully associative mapped cache with block size 4 KB. The size of main memory is 16 GB. Find the number of bits in tag.	8
9		Discuss briefly various pipelining hazards.	8
10		Define Buses. Discuss its types.	8
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